

Richwell Cyrille Santos Perez

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EDUCATION

University of Illinois Urbana-Champaign

August 2018 - May 2023

Master of Computer Science

GPA: 3.52

Bachelor of Science in Computer Science with Honors

GPA: 3.81

EXPERIENCE

RAVE Aerospace (formerly Safran Passenger Innovations)

Laramie, WY; Brea, CA

AI Engineer

June 2025 - Present

- Engineered a scalable RAG chatbot with Vue.js, Quart, and PostgreSQL, hosted on Azure App Services to serve 600+ internal users with managed identity authentication, async connection pooling, and structured logging for observability
- Architected a multi-agent LangGraph system with hierarchical supervisor routing queries to specialized SQL and knowledge agents, implementing 16 custom LangChain tools and dual PostgreSQL databases for operational MDM data and vector search
- Reduced query response latency by 50% through multi-agent routing optimization, prompt engineering, custom LangChain tools, and SSE-based streaming architecture
- Created a conversation analytics pipeline using K-Means/DBSCAN clustering and TF-IDF vectorization with UMAP/t-SNE visualization, identifying failure modes and optimizing agent prompts to improve response accuracy
- Established LangSmith evaluation harness with 6 evaluators across 40+ test cases, achieving 80%+ pytest coverage with SQL fact-checking against ground truth and JSONB reasoning capture for debugging
- Built a log parsing pipeline to process billions of records using Drain3 and PySpark, extracting ML features into Delta Lake
- Developing an LLM-powered log analyzer and an ML-based predictive maintenance system to forecast resets in IFE systems

Illinois Secretary of State

Springfield, IL

Database Administrator

February 2025 - June 2025

- Managed high-throughput DB2 databases (5 B+ records) on z/OS mainframe systems, optimizing queries and schemas to support statewide applications
- Improved query performance by designing and tuning complex SQL queries and schemas, reducing lookup time from 20 seconds to instantaneous for critical backend application queries
- Implemented ETL pipelines and automation scripts to streamline data migration and significantly reduce manual effort
- Collaborated with cross-functional development teams to design efficient schemas, stored procedures, and to integrate Azure-based services for scalable, cloud-connected applications
- Supported the development and deployment of statewide initiatives, including Apple mobile driver's licenses and REAL ID

University of Illinois Urbana-Champaign

Urbana, IL

Graduate Teaching Assistant

August 2022 - May 2023

- Led discussions and labs for Software Design & Database Systems, teaching design patterns and databases to 800+ students

Software Engineer

July 2021 - August 2021

- Developed and deployed LabWindows/CVI software to support new magnet mapping hardware, implementing real-time data interpretation, control logic, and optimizing performance

PROJECTS

UIUC Letter Grades — Full-Stack Web App with AI Integration

- Built a full-stack analytics platform with Node.js, MongoDB, and Python to visualize and analyze GPA data across 10+ years, enabling students to compare instructors and predict course outcomes
- Trained ML models (linear/logistic regression, sequential models) and visualizations for predictive grade analytics
- Analyzed and designed interactive data visualizations for users to track performance across semesters and predict future grades
- Implemented user authentication and automated data processing pipelines for efficient data handling

Video Anomaly Detection Model — Leveraging Weapon Recognition for Enhanced Accuracy

- Integrated weapon detection with anomaly scores, improving crime classification accuracy through multi-model decision-making
- Developed an AI pipeline using UCF-Crime anomaly detection & YOLO/Faster R-CNN for real-time crime recognition
- Optimized anomaly detection using Multi-Instance Learning (MIL) & improved loss functions for better predictions

Deep Learning Aerosol Model — 3rd Place, Ashby Prize in Computational Science, NCSA AI Innovation Hackathon

- Engineered a transformer and Encoder/Decoder models to capture temporal patterns in climate-relevant aerosol properties
- Applied advanced data preprocessing techniques to handle high-dimensional climate data and reduce the curse of dimensionality
- Leveraged HAL 9000 (UIUC's supercomputer) for high-performance training, optimizing deep learning workload scalability

SKILLS

Languages and Software: Python, Java, C/C++, JavaScript, SQL, MongoDB, PostgreSQL, Neo4j, Bash/Shell

AI/ML: TensorFlow, PyTorch, PySpark, Scikit-learn, LangChain, OpenCV, NLTK, Pandas, NumPy, Matplotlib

Frameworks and Tools: GCP, Azure, AWS, React.js, Node.js, Git, CI/CD, Docker, Postman, Microsoft Fabric, Delta Lake